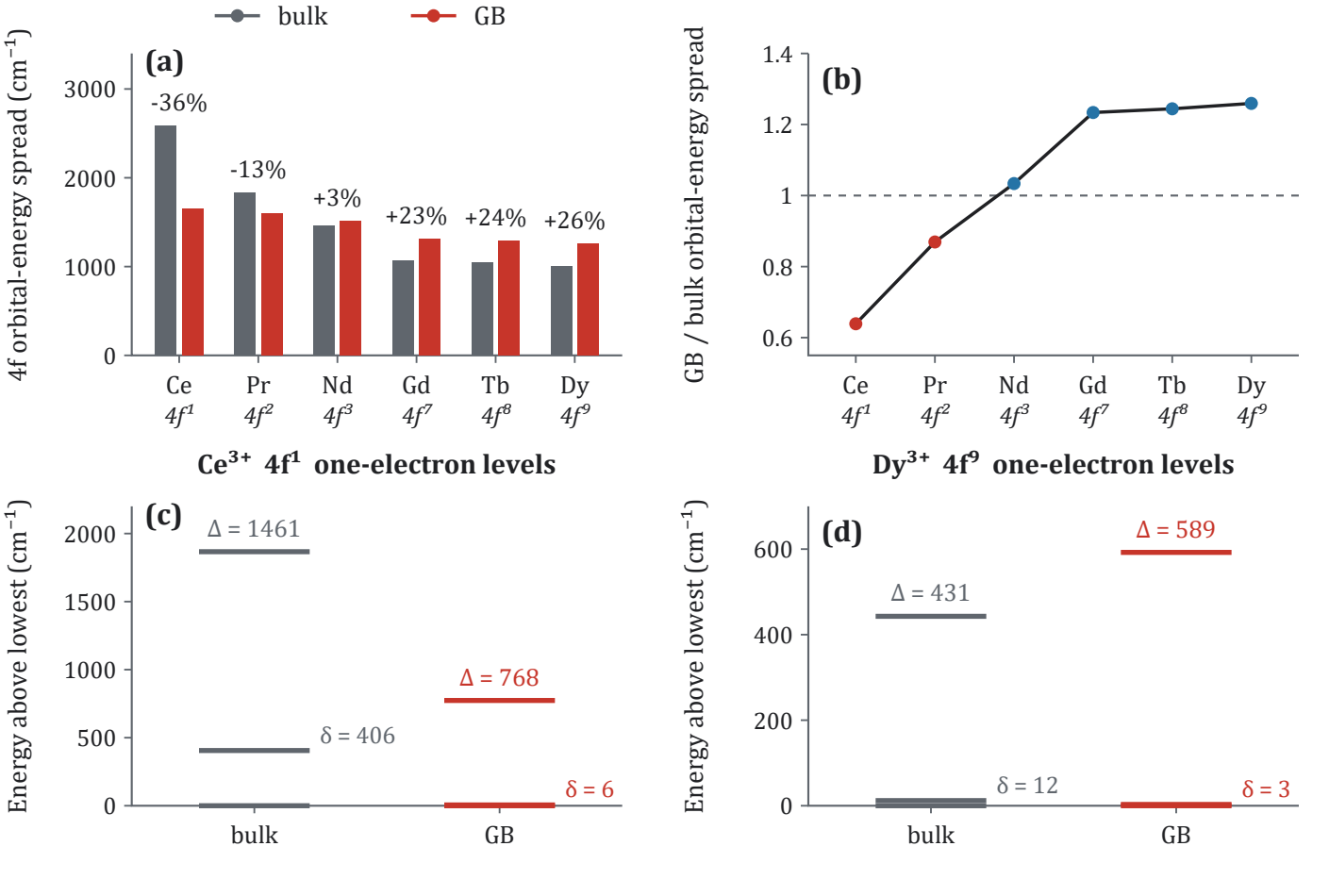

bulk
GB
(a)
4f orbital-energy spread (cm^{-1})
3000
2000
1000
0
-36%
-13%
+3%
+23%
+24%
+26%
Ce 4f^1
Pr 4f^2
Nd 4f^3
Gd 4f^7
Tb 4f^8
Dy 4f^9
Ce^{3+} 4f^1 one-electron levels
(b)
GB / bulk orbital-energy spread
1.4
1.2
1
0.8
0.6
Ce 4f^1
Pr 4f^2
Nd 4f^3
Gd 4f^7
Tb 4f^8
Dy 4f^9
Dy^{3+} 4f^9 one-electron levels
(c)
Energy above lowest (cm^{-1})
2000
1500
1000
500
0
Δ = 1461
δ = 406
Δ = 768
δ = 6
bulk
GB
(d)
Energy above lowest (cm^{-1})
600
400
200
0
Δ = 589
Δ = 431
δ = 12
δ = 3
bulk
GB